CSCI 441 - Software Engineering

SuperAsk.AI: AI-Powered Government Service Assistant

Report One

Team AO3

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Project URL: https://github.com/Monunhh/Super-ask.git

Table of Contents

SuperAsk.AI: AI-Powered Government Service Assistant.....1

1. Customer Problem Statement..... 3

2. Goals, System Requirements, and Analysis..... 7

3. Functional Requirement Specification and Use Cases..... 15

4. User Interface Specification.....34

5. Project Management..... 41

References.....46

Individual Contributions Breakdown

Team Member	Contribution to Report One, Part One
Yean Fong	Contributed to writing Section 2: Goals, System Requirements, and Analysis; captured prototype screenshots and integrated/formatted the final report. Contributed to writing Section 3b: Actors and Goals; prepared Section 5: Project Management, Gantt chart, team responsibilities, and references.
Doungkhe Neang	Contributed to writing Section 1: decomposition into sub-problems, glossary, and government-service content research; Section 3b: Actors and Goals.
Sereyprophea Sarin	Contributed to writing and reviewing Section 1: Customer Problem Statement; assisted with documentation, editing, requirement review, and QA-related content; Section 3a: Stakeholders.
Phallamonunhh Chun	Contributed to the technical prototype, including AI/RAG-related functionality and technical input supporting the system requirements; Section 3c-i: Casual Descriptions; Section 3c-ii: Use Case Diagram; Section 3c-iv: Fully-Dressed Descriptions.

Bunkheang Heng	Contributed to development of the technical prototype, including web/API, system architecture, database, and integration-related work; Section 3c-iii: Traceability Matrix; Section 3d: System Sequence Diagrams.
Heejae Choi	Contributed to UX/UI design and development of the prototype interface used to illustrate the User Interface Requirements; Section 4a: Preliminary design and Section 4b: User Effort Estimation.

1. Customer Problem Statement

Problem Statement

Scenario 1: High School Student Graduate

Right after completing my BACII National Examination, I wanted to move quickly and apply for university. I went to the application site to check the required documents, applying to both national and international universities at once. Across every application, the same two documents kept coming up: my high school diploma and my BACII results. For the results, I used the cropped version posted by the Ministry of Education on Facebook, since that was the only version available to me at the time.

After I submitted my application, I was told the documents were incorrect and that I needed to contact the ministry for verification. That instruction alone created a new problem, since I had no idea which ministry to go to. There are multiple sub-branches, each with a different location, and nothing on the application told me which one was responsible for verification.

I considered using an agency to handle the process for me, but backed out after hearing from friends that the full service costs $500, with 50 percent required upfront as a deposit before I even receive an acceptance letter. On top of the cost, I was also genuinely intimidated by the idea of navigating a government building on my own.

So I went to my teachers at the public school for guidance. They told me to go to my khan, but couldn't tell me where the office was or which specific one I needed. I then went to a general government office to ask directly, and they referred me to several different locations spread across the city. I followed their instructions one by one, and by the time I reached what was supposedly the correct office, I still didn't have the right documents with me. That entire day was lost.

Once I finally gathered the correct documents and returned, I learned about an additional fee and a late fee, and that my diploma also needed to be officially notarized before it would be accepted. Altogether, the process required layered approval from multiple separate offices and ministries, each with its own deadline and its own late fee structure, none of which were listed together anywhere. The combined waiting time across all these steps ended up being so long that I missed my application deadline entirely, despite starting the process right after my exam results came out.

The core problem throughout was that the information I needed was never in one place. It was scattered across different ministry websites, social media posts, and word of mouth, with no single source telling me the required documents, the correct offices, the fees, or the deadlines all at once.

Scenario 2: First-time Mothers

After giving birth to my son, I spent a week resting in the hospital. Before discharge, the doctors handed me a detailed medical report on my son and told me to get him registered. That was the extent of the guidance; I wasn't told which ministry or office was actually responsible for processing it.

Not wanting to spend hours checking every possible ministry, I decided to set aside time and figure it out myself. But time was already tight: I was on pregnancy leave, though still expected to do remote work as an assistant to my team; I was physically recovering from labour, and I was hosting daily visits from family coming to see the new baby. Realistically, I could only carve out one full day to handle the registration in person.

Weekends were my only free window, so that's when I tried to go. Each time, something didn't line up: the office's shift had ended, it was their lunch break, or it simply wasn't a working day for them. My free time and their available hours never overlapped. Between recovering from childbirth, caring for the baby alone, and managing remote work, I couldn't shift my schedule to match theirs, and neither could they shift to match mine. Two weeks passed this way, with every attempt missing.

A month in, I finally got a Monday off, a weekday, when the office would actually be open. I brought the documents the hospital gave me and went to register. Only then did I learn several things at once: I had already missed the registration deadline, incurring late fees on top of the original fee; the hospital's report wasn't sufficient on its own, since I needed additional specific documents; and I was at the wrong office entirely, since I needed to go through my khan, not the central ministry.

So a month of effort, plus money spent on the fee and now a late penalty, resulted in nothing being filed. And I still don't know exactly which documents I need to bring or how they need to be formatted so that next time, I can submit everything correctly in one visit, without having to rely on informally checking Telegram and Facebook updates from both the ministry and the khan just to stay current, on top of whatever unspoken procedural rules aren't posted anywhere at all.

Scenario 3: National Election Registration

Election season came around, and I knew I needed to register to vote. The problem was, the deadline wasn't sent to me directly. It was only announced through the Ministry's Telegram channel and Facebook page. So I had to go check for myself, scrolling through weeks of posts to find the actual date, because the feed was flooded with unrelated announcements, reposts, and comments, and I couldn't always tell which post was the most recent or which one had been quietly updated or corrected later. I bookmarked what I thought was the right date, but I had no way of confirming whether it was still accurate without checking again and again.

On top of that, I wasn't sure if my ID was even valid for registration anymore. Nobody tells you outright; you find out the procedure depends on your age group. Younger first-time voters go through one process, people renewing an expired ID go through another, and everyone assumes you already know which category you fall into. I didn't.

Here's the part that confused me most: registering to vote itself is free. But if my ID happens to be outdated, I get charged a separate fee to renew a fee that has nothing to do with the election, but that I have to pay anyway just to be eligible. And if I miss that ID renewal window, I don't just pay a penalty; I lose the ability to vote entirely, with no appeal and no second window. So a paperwork issue that's technically unrelated to the election becomes the single point of failure that decides whether I get a say at all.

I ended up spending more time trying to verify whether my information was current than actually completing the registration.

Scenario 4: Government Officers

From the other side of the counter, the picture looks different, but the outcome is the same. The office is short-staffed, and the people who are there are working shifts because they're human beings who also need lunch breaks, rest, and time off. When a citizen shows up expecting help exactly when the office happens to be short-handed or between shifts, there's often no one immediately free to assist, not because no one cares, but because there simply aren't enough people to cover every hour without gaps.

Meanwhile, the staff who are on duty are constantly interrupted. Someone walks in with a question, another person cuts the line, and a third shows up with the wrong documents entirely. Each interruption breaks whatever task the officer was in the middle of completing. Multiply that by dozens of people a day, and very little progress is made on any single case.

A lot of the workload is also self-inflicted by the process itself: citizens frequently arrive with documents that aren't formatted the way the system requires, missing the specific keywords or fields needed for official filing. The officer then has to explain what's wrong, send the person away to fix it, and wait for them to come back. Sometimes days or weeks later, restarting the same conversation from scratch. This repetitive back-and-forth eats up time that could otherwise go toward processing new cases.

And because the office doesn't have the staffing to individually track every applicant's status, there's no consistent way to flag "this person is close to a deadline" or "this person already tried once and got turned away." Each visit is treated as if it's the first, because there isn't the manpower to do otherwise. It would be a miracle if the citizens were able to access a sample and know what is required of them beforehand so when they come, the document can be registered right away, saving us both the overhead.

Decomposition into Sub-problems

High school graduates

- Identify the procedure for getting school documents notarized or certified.

- Determine what documents are required, such as the high school diploma and BACII results.
- Explain where students need to go and the steps they need to follow.
- Help students understand what documents are accepted when official results or certificates are not yet available.

First-time parents

- Explain how to register a newborn and obtain a birth certificate.
- Identify the required documents.
- Provide information about registration deadlines, locations, and possible fees.

Citizens registering for elections

- Provide information about voter registration deadlines.
- Explain what documents are required and whether they need to be renewed.
- Identify where citizens need to register and whether there are any fees.

Government officers

- Reduce repetitive questions and tasks handled by government staff.
- Organize commonly requested information so citizens can access it themselves.
- Reduce workload and make the process of providing government information more efficient.

Glossary of Terms

Terms	Meaning
BacII	Cambodia's National Exam.
Notarize	To officially verify that a document or signature is real.
Government Contact Referral	The process of directing a citizen to the relevant government ministry or office by providing available official contact information and location when additional assistance from the government office is needed.
Unsupported Query	A question that SuperAsk cannot answer because there is no approved government information available in its database.
Effective Date	The date when a law, rule, procedure, or requirement officially begins to apply.
Last-Verified Date	The most recent date when information in SuperAsk was checked and confirmed to still be accurate.

Legal Instrument	An official legal document, such as a law, sub-decree, proclamation, regulation, or directive, that provides authority for a government procedure.
Government Procedure	The official steps a citizen must follow to access or complete a government service.
Khan	Administrative district in Cambodia

2. Goals, System Requirements, and Analysis

Goals

The business goals below show how SuperAsk.AI's main objective is divided into four supporting goals: comprehensive government-service information, trustworthy answers, bilingual accessibility, and ministry contact and location information.

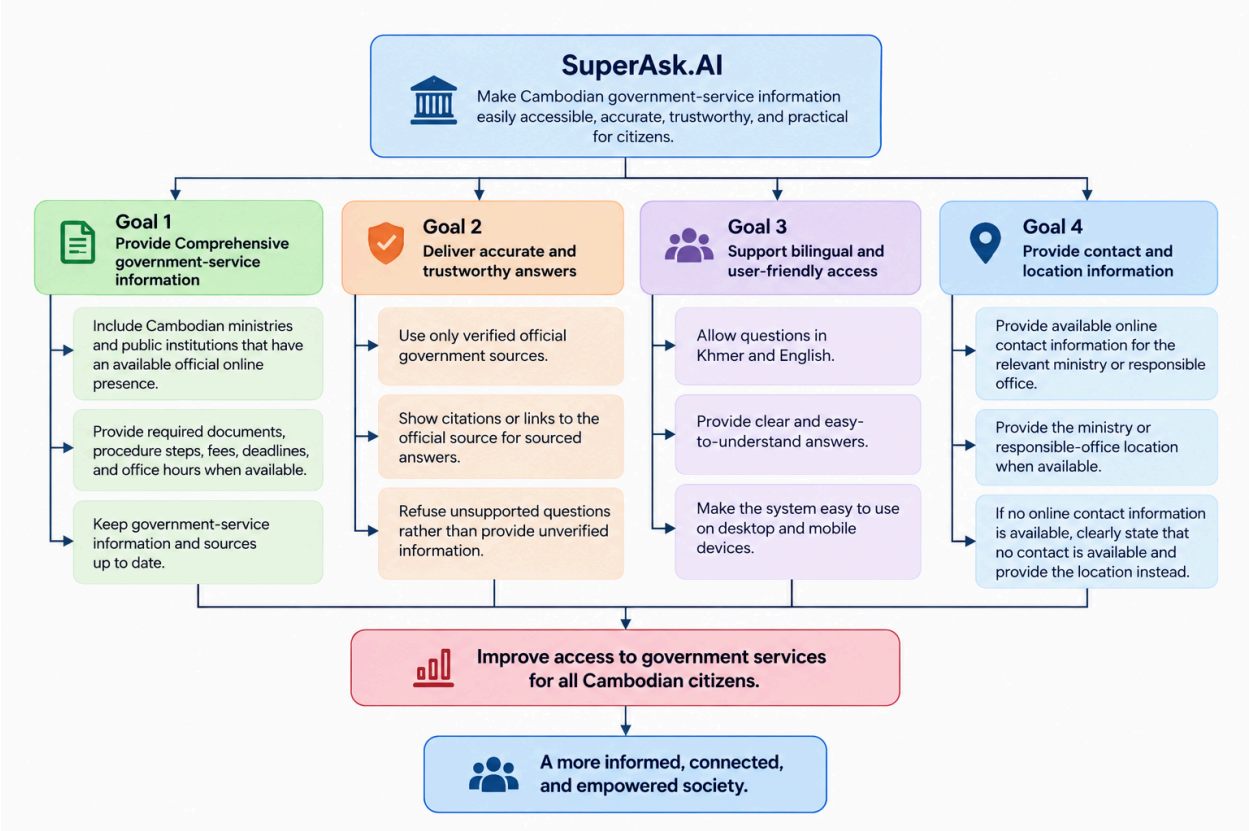


Figure 1: Project Goals

Enumerated Functional Requirements

Priority Weight (PW) uses a scale from 1 to 5, where 5 represents the highest-priority requirements and 1 represents the lowest-priority or optional requirements.

Identifier	PW	Requirement
REQ-1	5	The system shall provide a chat interface where users can submit questions about Cambodian government services.
REQ-2	5	The system shall accept questions in Khmer and English.
REQ-3	4	The system should support mixed Khmer-English questions.
REQ-4	5	The system shall retrieve relevant information from verified official government online sources before generating an answer.
REQ-5	5	The system shall generate answers using only relevant information retrieved from approved government sources.
REQ-6	5	The system shall display the official source or citation supporting each sourced answer.
REQ-7	5	The system shall decline to provide an unsupported answer when no verified official source can be found.
REQ-8	5	The system shall provide government-service information for Cambodian ministries and public institutions that have an available official online presence.
REQ-9	4	The system shall allow users to browse or select commonly requested government procedures.
REQ-10	3	The system shall display recent questions or conversations to help users return to previously viewed topics.
REQ-11	4	The system shall provide available online contact information for the relevant ministry or responsible office when such information is available from an official source.
REQ-12	4	The system shall provide the location of the relevant ministry or responsible office when location information is available from an official source.
REQ-13	4	If no online contact information is available, the system shall inform the user that no online contact is available and provide the ministry or responsible-office location instead.

REQ-14	4	The system shall maintain conversation context so users can ask follow-up questions during the same session.
REQ-15	3	The system should support access through the web, Telegram, and Messenger.
REQ-16	5	The system shall provide the required documents for a government procedure when the information is available from a verified official source.
REQ-17	5	The system shall provide the steps required to complete a government procedure in the correct order when available from a verified official source.
REQ-18	5	The system shall provide official fees and applicable late fees when available from a verified official source.
REQ-19	5	The system shall provide applicable deadlines or validity periods when available from a verified official source.
REQ-20	4	The system shall provide available office hours and other relevant office information when provided by a verified official source.

Enumerated Nonfunctional Requirements

The numbering continues from the functional requirements.

Identifier	PW	Requirement
REQ-21	5	At least **90% of test users** should be able to submit a government-service question through the main chat interface without assistance.
REQ-22	5	During acceptance testing, no sourced answer shall contain a factual claim that is unsupported by its displayed verified official source.
REQ-23	5	The system shall correctly refuse at least **95% of unsupported or out-of-scope questions** included in the project test set.
REQ-24	4	Under normal testing conditions, the system should begin displaying a response within **5 seconds for at least 90% of test queries**.
REQ-25	4	The web interface shall remain usable without horizontal scrolling on screens from approximately **360 pixels wide** through standard desktop resolutions.

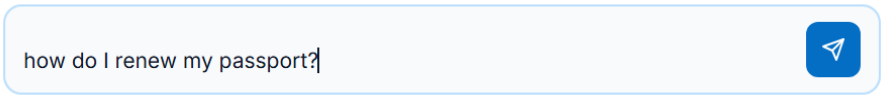
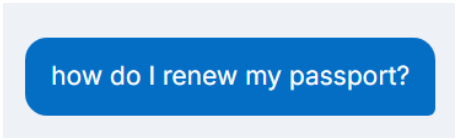
REQ-26	4	Khmer and English text shall display without broken, missing, or unreadable characters in the browsers used for project testing.
REQ-27	5	API keys, model credentials, and other system secrets shall remain server-side and shall not be accessible through the citizen-facing browser interface.
REQ-28	4	Every sourced answer shall visually separate the answer content from its official source or citation information.
REQ-29	4	Authorized government content and ministry information shall be updateable through stored data or configuration without requiring changes to the core answer-generation logic.
REQ-30	4	Additional ministries and public institutions shall be addable without redesigning the core retrieval and answer-generation architecture.

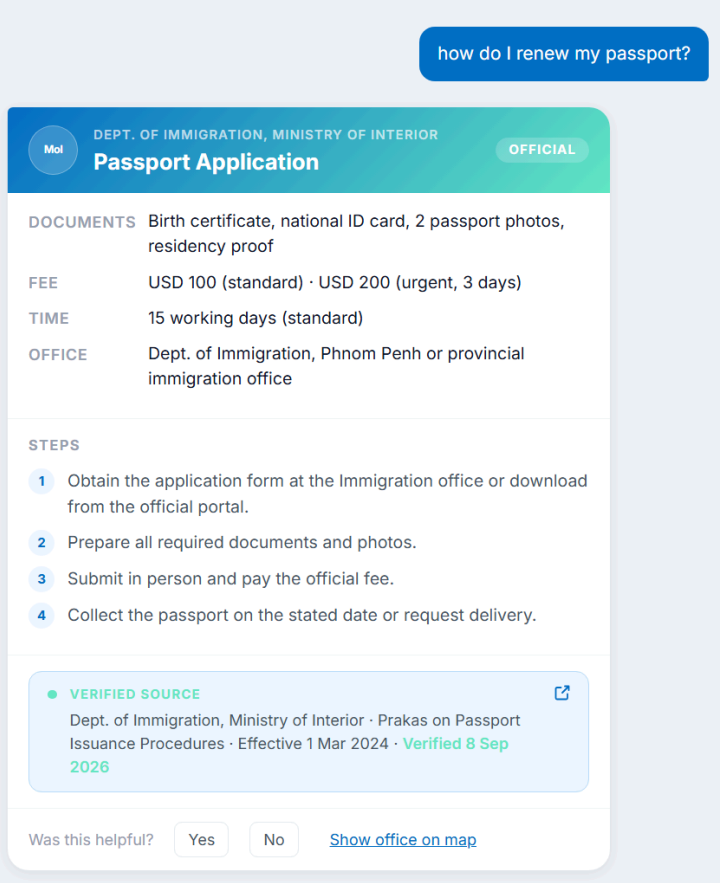
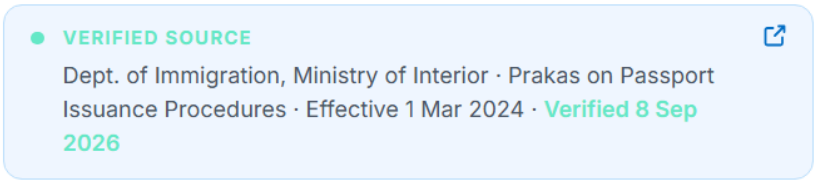
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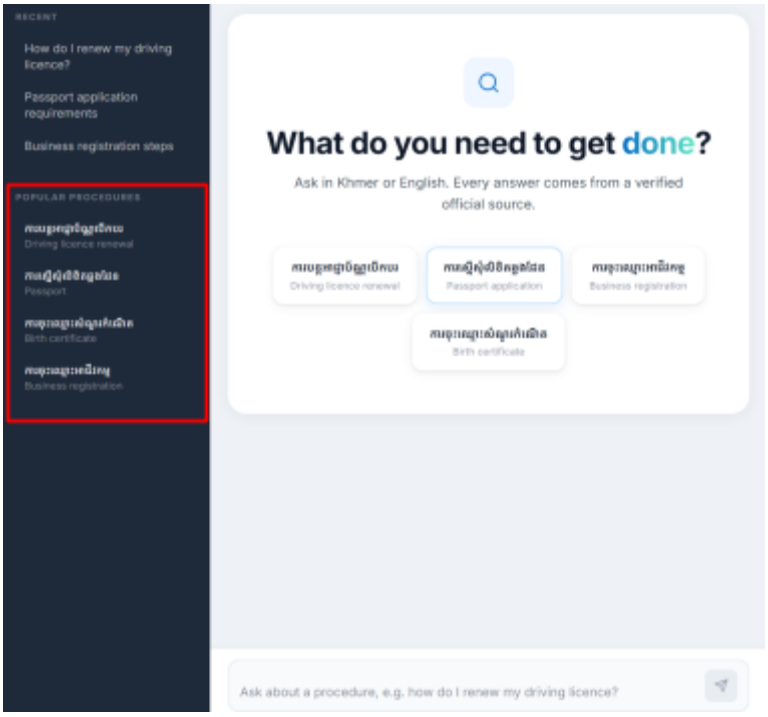
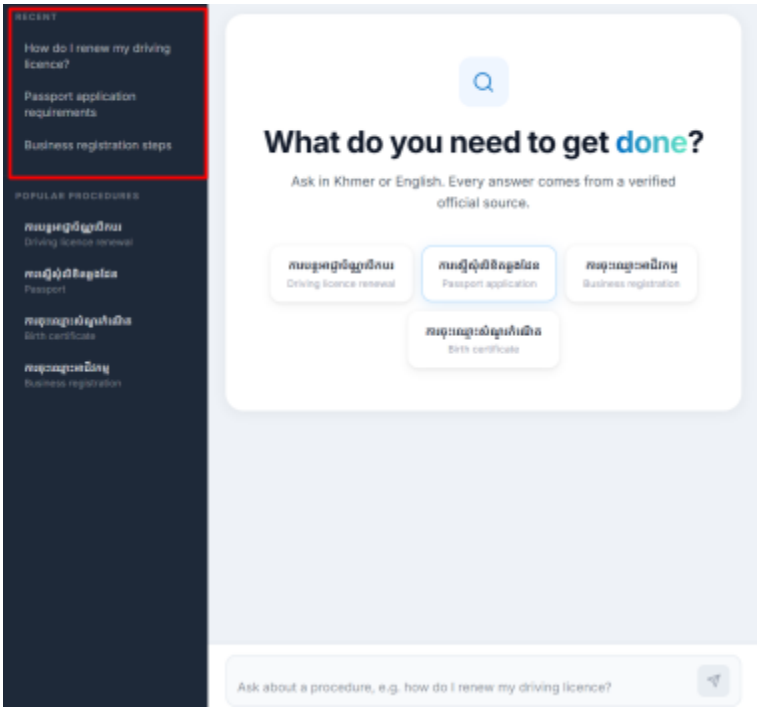
FURPS+ Category	SuperAsk.AI
Functionality	The system supports government-service questions, Khmer and English interaction, verified-source retrieval, citations, procedure steps, required documents, fees, deadlines, office information, ministry contact details, and ministry locations.
Usability	The chat interface should be simple, clearly organized, responsive, and usable on both mobile and desktop devices. At least 90% of test users should be able to submit a question without assistance.
Reliability	The system should answer using verified official sources and refuse unsupported questions rather than guess. Sourced answers must remain consistent with their displayed sources during acceptance testing.
Performance	Under normal testing conditions, the system should begin displaying a response within 5 seconds for at least 90% of test queries.
Supportability	Government information, ministry contacts, office locations, and official sources should be updateable without changing the core answer-generation logic. Additional ministries should also be addable without major architectural redesign.

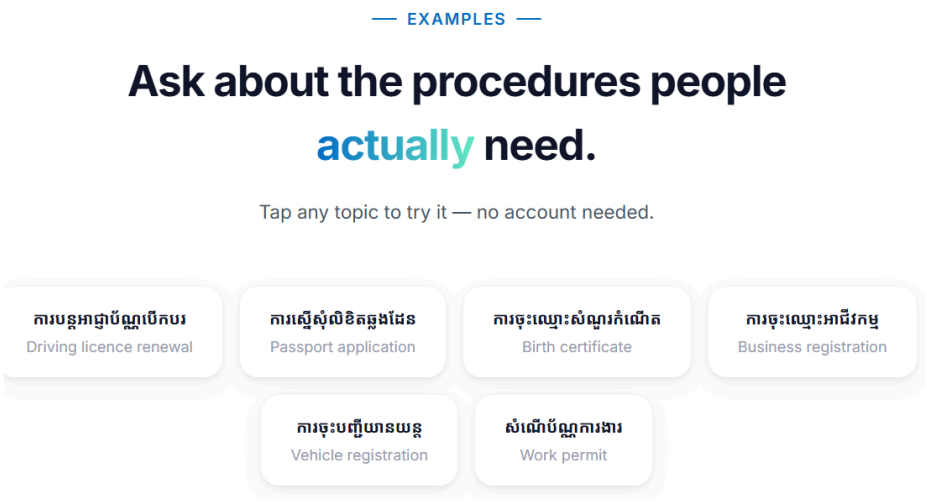
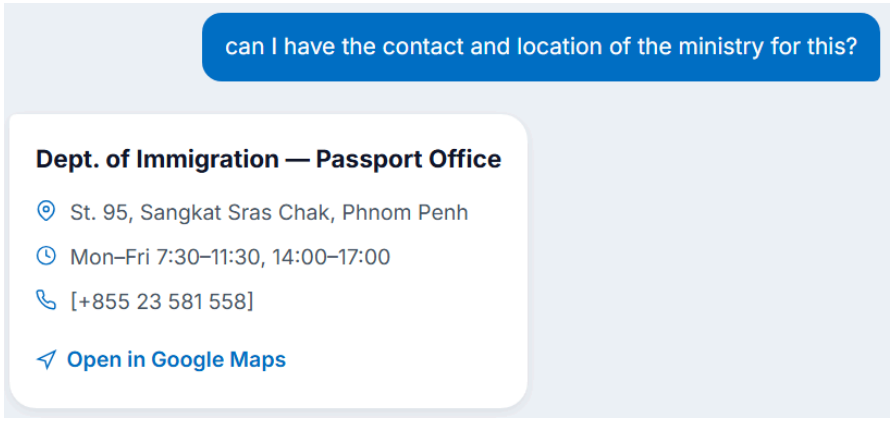
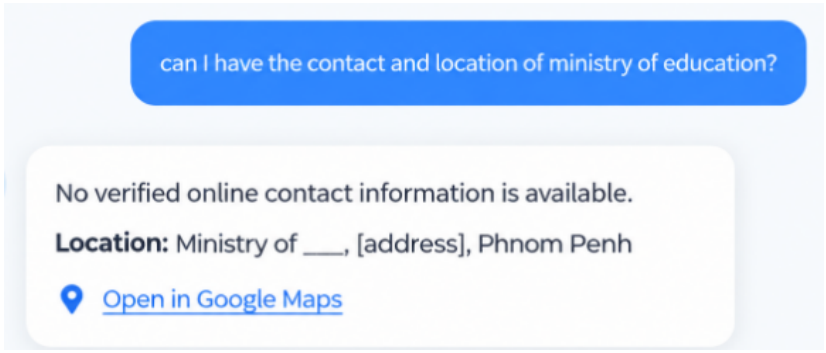
Security (+)	API credentials, model keys, and other system secrets must remain server-side and inaccessible through the citizen-facing browser interface.
Design / Implementation Constraints (+)	SuperAsk.AI shall operate as a web-based service, use verified official government information for sourced answers, and support responsive access on both mobile and desktop devices.

User Interface Requirements

Identifier	Priority	Requirement
REQ-31	5	A user shall be able to enter a government-service question in the chat input field. 
REQ-32	5	A user shall be able to submit a question to SuperAsk.AI. 
REQ-33	5	A user shall be able to view the answer generated by SuperAsk.AI.

		
REQ-34	5	A user shall be able to view the verified official source or citation supporting the answer. 
REQ-35	4	A user shall be able to select a popular procedure from the interface.

		
REQ-36	4	A user shall be able to view recent questions or conversations. 
REQ-37	4	A user shall be able to use the interface in Khmer or English.

		
REQ-38	4	A user shall be able to view available ministry contact and location information when provided by an official source. 
REQ-39	4	If no online contact information is available, the interface shall state that no contact is available and display the ministry or responsible-office location. 

3. Functional Requirement Specification and Use Cases

a. Stakeholders

Having a reliable, centralized way to access accurate government service information can reduce the confusion, wasted trips, and missed deadlines that currently push citizens toward costly agencies or repeated failed visits to the wrong office. This can also ease pressure on government offices themselves, since fewer citizens will arrive with incomplete documents or outdated information. Making government procedures easier to understand and follow benefits individual citizens as well as the broader public institutions they depend on. However, we have identified these primary stakeholders:

Citizens (Individual Users): use the system to look up information about government procedures such as document notarization, birth registration, and voter registration. These users will benefit by having accurate, up-to-date requirements, deadlines, and fees available in one place instead of scattered across ministry websites, Facebook posts, and word of mouth. Having a bilingual chat interface that clearly cites its sources will allow citizens to prepare correctly before visiting an office, reducing the stress, wasted time, and unexpected fees described in Scenarios 1 through 3.

Government Officers and Public Institutions: Do not use SuperAsk.AI directly, but they publish the official information it relies on and remain the point of contact for citizens who prepare through the app. These stakeholders will benefit by receiving fewer repetitive questions and fewer citizens arriving with missing or incorrectly formatted documents. Reducing this repetitive back-and-forth, as described in Scenario 4, will free up staff time to process more cases and reduce the strain caused by understaffing and constant interruptions.

Ministries and Government Agencies: These are the organizations whose official procedures, requirements, and contact information are published on their websites, Facebook pages, and Telegram channels, and collected into the system from there. These stakeholders have an interest in ensuring the information SuperAsk.AI displays is accurate, properly sourced, and kept current, since any errors would reflect on the ministry itself. As more ministries are added under REQ-30, this group will continue to grow alongside the system's coverage.

Professor Hieu Vu and the CSCI 441 Course serve as the academic sponsor and evaluator of the project. This stakeholder has an interest in seeing the team apply sound software engineering practices throughout requirements gathering, design, implementation, and testing, and in reviewing the team's progress at each report and demonstration milestone.

Team AO3 (Developers): are deeply invested in the success of SuperAsk.AI. Each member is responsible for a specific area, including AI and retrieval accuracy, backend and database development, content curation, quality assurance, and interface design. The team's dedication to building a functional, trustworthy, and easy to use system is critical to ensuring the project meets both its academic requirements and its intended real world purpose.

b. **Actors and Goals**

Actor	**Participating / Initiating**	**Role**	**Goal**
Citizen / User	Initiating	The User interacts with SuperAsk.AI to ask questions about Cambodian government procedures and services.	The goal of the User is to receive accurate government-service information in Khmer or English, including procedures, required documents, fees, deadlines, office information, and verified sources.
AI Model Service	Participating	The AI Model Service processes the user's question together with retrieved government information to generate a relevant response.	The goal of the AI Model Service is to generate a clear response based on the approved information provided by SuperAsk.AI.
Telegram Platform	Participating	The Telegram Platform provides an additional communication channel through which users may interact with SuperAsk.AI.	The goal of the Telegram Platform is to transmit user questions to SuperAsk.AI and deliver generated responses back to the user.
Messenger Platform	Participating	The Messenger Platform provides an additional communication channel through which users may interact with SuperAsk.AI.	The goal of the Messenger Platform is to transmit user questions to SuperAsk.AI and deliver generated responses back to the user.
Scheduler (System Timer)	Initiating	The Scheduler starts SuperAsk.AI's content refresh at set intervals, without any user action.	The goal of the Scheduler is to keep the stored government information current, so that answers and last-verified dates reflect what ministries have most recently published.
Official Sources (Ministry Websites, Facebook Pages, Telegram Channels)	Participating	The Official Sources are the public online channels where ministries publish procedures, fees, deadlines, and contact details. SuperAsk.AI crawls them to collect its content.	The goal of the Official Sources is to provide the up-to-date official information that every SuperAsk.AI answer and citation is based on.

c. Use Cases

3c-i. Casual Description

The use cases below cover every system requirement from Report One, Part One. Each one lists the requirements it responds to. UC-1 to UC-10 are started by the Citizen / User. UC-11 is started by the system's own scheduler. It keeps the stored government information current by crawling official ministry websites, Facebook pages, and Telegram channels.

Use Case	Name	Casual Description	Requirements
UC-1	AskQuestion	The user types a question about a Cambodian government service in Khmer, English, or a mix of both, and submits it. SuperAsk.AI finds the relevant official information and returns an answer. Where the source has them, the answer lists the procedure steps in order, the required documents, fees and late fees, deadlines or validity periods, and office hours.	REQ-1, REQ-2, REQ-3, REQ-5, REQ-16, REQ-17, REQ-18, REQ-19, REQ-20, REQ-31, REQ-32, REQ-33
UC-2	RetrieveOfficialSources	The system searches its stored official government content for the passages most relevant to the question and passes only those to the AI Model Service. It is always included in UC-1.	REQ-4, REQ-5, REQ-8
UC-3	ViewSourceCitation	The user sees the official source behind the answer: the ministry, the document or page, a link, the effective date, and the last-verified date. The citation is kept visually separate from the answer text. It is always included when a sourced answer is shown.	REQ-6, REQ-28, REQ-34
UC-4	DeclineUnsupportedQuestion	If no verified official source covers the question, or the question is out of scope, the system says it cannot answer. It does not guess. It tells the user which areas it does cover so they can rephrase. This extends UC-1.	REQ-7
UC-5	BrowsePopularProcedures	The user picks a common procedure, such as passport application or birth certificate, from the sidebar or the welcome screen. The system sends it as a question and answers it as in UC-1.	REQ-9, REQ-35

UC-6	ViewRecentConversations	The user opens the list of recent questions and conversations and selects one to go back to a topic they viewed earlier.	REQ-10, REQ-36
UC-7	GetOfficeContactAndLocation	The user asks for the contact details and location of the ministry or office that handles their procedure. The system shows the phone number, address, office hours, and a map link. If there is no online contact, it says so and shows the office location instead.	REQ-11, REQ-12, REQ-13, REQ-20, REQ-38, REQ-39
UC-8	AskFollowUpQuestion	After receiving an answer, the user asks a follow-up question, such as "how much does it cost?". The system uses the earlier conversation to work out what "it" refers to. This extends UC-1.	REQ-14
UC-9	AskThroughMessagingChannel	The user sends a question to SuperAsk.AI through Telegram or Messenger instead of the website. The platform forwards it, the system answers it through UC-1, and the platform sends back the answer with its source.	REQ-15
UC-10	SwitchLanguage	The user switches the interface between Khmer and English. Answers come back in the language the question was asked in.	REQ-2, REQ-37
UC-11	RefreshOfficialContent	On a schedule, the system crawls the registered official sources: ministry websites, Facebook pages, and Telegram channels. It saves new or changed content with its source link and fetch date, and updates the stored information and its last-verified date. New ministries are added by registering their sources, without changing the answer-generation logic.	REQ-4, REQ-8, REQ-29, REQ-30

3c-ii. Use Case Diagram

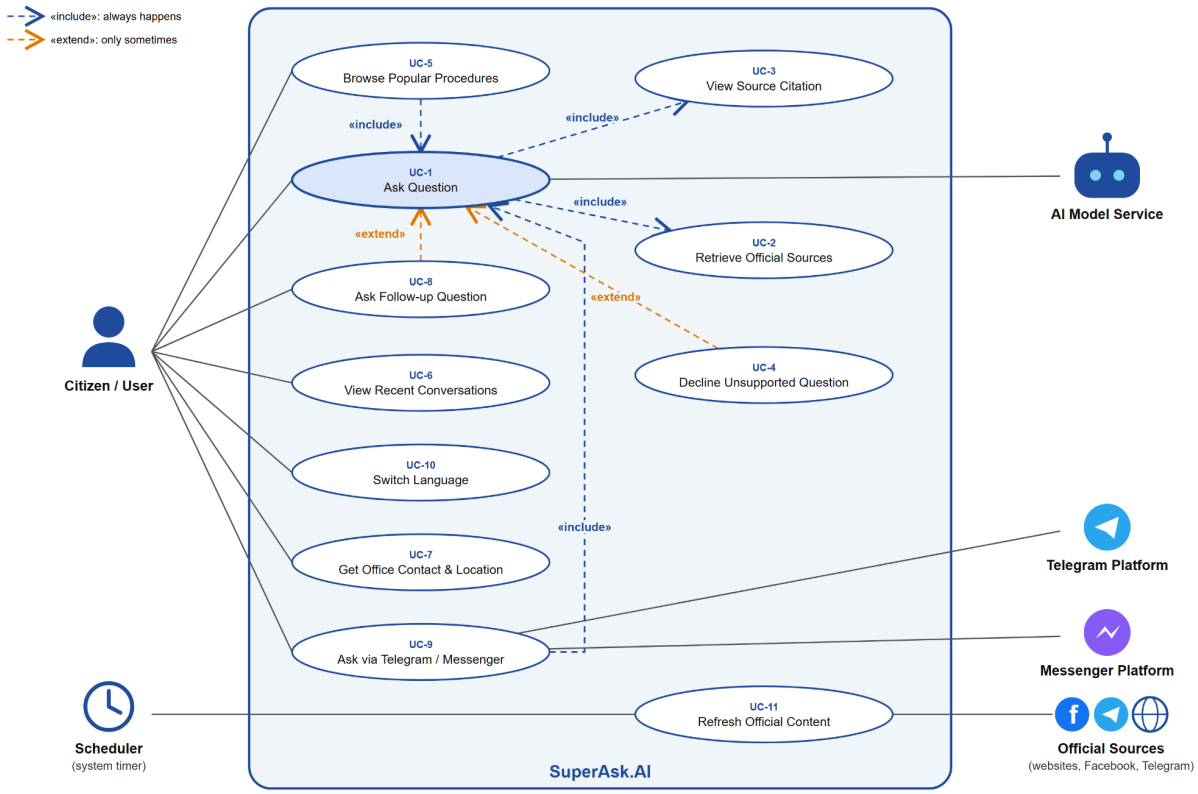


Figure 2: Use Case Diagram

The diagram above shows how the SuperAsk.AI use cases relate to each other and to the actors. AskQuestion (UC-1) is the central use case: it always includes RetrieveOfficialSources (UC-2) and ViewSourceCitation (UC-3), because every answer is looked up and cited. BrowsePopularProcedures (UC-5) and AskThroughMessagingChannel (UC-9) include AskQuestion, since they are other ways of asking the same question. AskFollowUpQuestion (UC-8) and DeclineUnsupportedQuestion (UC-4) extend AskQuestion, because they only happen in some conversations. RefreshOfficialContent (UC-11) is started by the Scheduler and collects content from the official ministry websites, Facebook pages, and Telegram channels.

3c-iii. Traceability Matrix (1) - System Requirements to Use Cases

The traceability matrix below maps each system requirement from Report One, Part One to the use cases that satisfy it. The priority weight of a use case is the sum of the priority weights (PW) of the requirements it covers. Nonfunctional requirements REQ-21 to REQ-27 (usability, accuracy, refusal rate, response time, screen width, Khmer display,

and security) are quality constraints on the whole system rather than on a single use case, so they are verified through system testing instead of being traced here.

Req't	PW	UC-1	UC-2	UC-3	UC-4	UC-5	UC-6	UC-7	UC-8	UC-9	UC-10	UC-11
REQ-1	5	X										
REQ-2	5	X									X	
REQ-3	4	X										
REQ-4	5		X									X
REQ-5	5	X	X									
REQ-6	5			X								
REQ-7	5				X							
REQ-8	5		X									X
REQ-9	4					X						
REQ-10	3						X					
REQ-11	4							X				
REQ-12	4							X				
REQ-13	4							X				
REQ-14	4								X			
REQ-15	3									X		
REQ-16	5	X										
REQ-17	5	X										
REQ-18	5	X										
REQ-19	5	X										
REQ-20	4	X						X				
REQ-28	4			X								
REQ-29	4											X

REQ-30	4											X
REQ-31	5	X										
REQ-32	5	X										
REQ-33	5	X										
REQ-34	5			X								
REQ-35	4					X						
REQ-36	4						X					
REQ-37	4										X	
REQ-38	4							X				
REQ-39	4							X				
Max PW		**58**	**15**	**14**	**5**	**8**	**7**	**24**	**4**	**3**	**9**	**18**

Ranked by priority weight: UC-1 (58), UC-7 (24), UC-11 (18), UC-2 (15), UC-3 (14), UC-10 (9), UC-5 (8), UC-6 (7), UC-4 (5), UC-8 (4), UC-9 (3). The five highest-weighted use cases, UC-1 AskQuestion, UC-7 GetOfficeContactAndLocation, UC-11 RefreshOfficialContent, UC-2 RetrieveOfficialSources, and UC-3 ViewSourceCitation, are elaborated in Sections 3c-iv and 3d and planned for the first demo. UC-4 DeclineUnsupportedQuestion has a low weight because it traces to a single requirement, but it will also be shown in the first demo, since it runs inside UC-1 whenever no verified source is found.

3c-iv. Fully-Dressed Description

The five use cases below have the highest priority weights in the traceability matrix (Section 3c-iii) and form the core of the first demonstration. In each flow, → marks an action by an actor and ← marks the system's response.

Use Case UC-1: AskQuestion	
Requirements	REQ-1, REQ-2, REQ-3, REQ-5, REQ-16, REQ-17, REQ-18, REQ-19, REQ-20, REQ-31, REQ-32, REQ-33

Initiating Actor	Citizen / User
Actor's Goal	To get accurate information about a government procedure (steps, required documents, fees, deadlines, office hours) in Khmer or English.
Participating Actors	AI Model Service
Preconditions	The SuperAsk.AI web interface is open. The stored official content has been loaded (UC-11 has run at least once).
Postconditions	The user has an answer supported by a displayed official source, or a clear statement that no verified source is available (UC-4). The question and answer appear in the recent conversations list.
Flow of Events for Main Success Scenario	→ 1. **User** types a question in Khmer, English, or a mix of both into the chat input field. → 2. **User** submits the question by pressing Enter or the send button. ← 3. **System** shows the question in the conversation and a loading indicator. ← 4. **System** detects the question's language and normalises the Khmer text. ← 5. **System** checks that the question is about a government service and not an out-of-scope request. ← 6. **include::** RetrieveOfficialSources (UC-2). ← 7. **System** sends the question and the retrieved passages to the **AI Model Service**, which writes an answer using only those passages. ← 8. **System** checks that every fee, date, and document in the answer appears in a cited source.

	← 9. **System** shows the answer in the question's language. Where available, it lists the steps in order, the required documents, fees and late fees, deadlines, and office hours. ← 10. **include::** ViewSourceCitation (UC-3). ← 11. **System** adds the question to the recent conversations list.
Flow of Events for Extensions (Alternate Scenarios)	2a. User submits an empty input. ← 1. System ignores the submission and keeps the send button disabled. 5a. The question is not about a government service (for example, general knowledge or translation). ← 1. System tells the user that SuperAsk.AI only answers questions about Cambodian government services, and makes no retrieval or model call. 6a. No passage relevant enough to the question is found. ← 1. **extend::** DeclineUnsupportedQuestion (UC-4). 7a. The AI Model Service is unavailable or times out. ← 1. System falls back to showing the most relevant retrieved passage word for word, with its citation. 8a. The answer contains a claim that no cited source supports. ← 1. The system withholds the answer and tells the user it could not confirm every detail against an official source. 9a. The user asks a follow-up question about the answer. → 1. **extend::** AskFollowUpQuestion (UC-8).

Use Case UC-2: RetrieveOfficialSources	
Related Requirements	REQ-4, REQ-5, REQ-8
Initiating Actor	Included by UC-1 AskQuestion (on behalf of Citizen / User)
Actor's Goal	To find the official passages that answer the question, so that the answer is based only on them.
Participating Actors	None (uses the system's stored official content)
Preconditions	A question has been submitted and passed the scope check. The content store contains official content collected by UC-11.
Postconditions	A ranked list of relevant passages is returned, each with its source metadata, or an empty result that triggers UC-4.
Flow of Events for Main Success Scenario	← 1. **System** splits the Khmer or English question into search terms and adds known synonyms and abbreviations (for example, "MPWT" and "Ministry of Public Works and Transport"). ← 2. **System** runs a keyword search and a meaning-based search over the stored content. ← 3. **System** merges and re-ranks the two result lists. ← 4. **System** removes outdated or replaced content, keeping only content that is currently in effect. ← 5. **System** checks that the top result scores above the minimum relevance threshold.

	← 6. **System** returns the top passages, each with its ministry, source link, effective date, and last-verified date.
Flow of Events for Extensions (Alternate Scenarios)	2a. The question is a follow-up that does not name its subject (for example, "how much is it?"). ← 1. System rewrites the question using the earlier conversation before searching. 5a. No result reaches the minimum relevance threshold. ← 1. System returns an empty result, and the calling use case continues with DeclineUnsupportedQuestion (UC-4). 5b. The top result is related but does not mention most of what was asked. ← 1. System marks the result as low confidence, so the answer is shown with a note that the source may not fully answer the question.

Use Case UC-3: ViewSourceCitation	
Related Requirements	REQ-6, REQ-28, REQ-34
Initiating Actor	Citizen / User
Actor's Goal	To confirm where the answer came from and how current it is.

Participating Actors	None
Preconditions	A sourced answer has been shown (UC-1).
Postconditions	The user has seen the source details and, if they chose to, opened the official source.
Flow of Events for Main Success Scenario	← 1. **System** shows a "Verified Source" card under the answer, visually separate from the answer text. ← 2. **System** shows the ministry, document or page title, effective date, and last-verified date on the card. → 3. **User** reads the source details. → 4. **User** selects the link icon on the card. ← 5. **System** opens the official source page (ministry website, Facebook post, or Telegram post) in a new tab.
Flow of Events for Extensions (Alternate Scenarios)	2a. The source's last-verified date is past its review date. ← 1. System marks the citation as possibly outdated and adds a note asking the user to confirm with the office before acting. 2b. The answer is supported by more than one source. ← 1. System shows one card per source, in the order they are cited. 3a. User finds the information incorrect. → 1. User selects "Report" on the answer and optionally adds a comment.

	← 2. System records the report against the cited source for review.

Use Case UC-7: GetOfficeContactAndLocation	
Related Requirements	REQ-11, REQ-12, REQ-13, REQ-20, REQ-38, REQ-39
Initiating Actor	Citizen / User
Actor's Goal	To know how to contact the responsible ministry or office and where to go in person.
Participating Actors	None (uses the stored office information)
Preconditions	The web interface is open. Contact and location details for the office have been collected from its official website, Facebook page, or Telegram channel (UC-11).
Postconditions	The user has the office's contact details and location, or a clear statement that no online contact is available together with the office location.

Flow of Events for Main Success Scenario	→ 1. **User** asks for the contact and location of the office responsible for a procedure (for example, "can I have the contact and location of the ministry for this?"). ← 2. **System** works out which procedure "this" refers to from the conversation (UC-8). ← 3. **System** finds the ministry or office responsible for that procedure. ← 4. **System** shows a contact card with the office name, address, office hours, phone number, and an "Open in Google Maps" link. ← 5. **System** shows the official source the contact details came from. → 6. **User** selects "Open in Google Maps". ← 7. **System** opens the office location in Google Maps in a new tab.
Flow of Events for Extensions (Alternate Scenarios)	4a. No online contact information is available for the office. ← 1. System states "No verified online contact information is available." ← 2. System shows the office location and the Google Maps link instead. 4b. Neither contact nor location information is available from an official source. ← 1. System says it has no verified contact or location for this office and does not guess one. 3a. The procedure is handled by several offices (for example, a provincial office and the ministry in Phnom Penh). ← 1. System asks the user which province they are in.

	→ 2. User answers, and the flow continues at step 4 with the matching office. 1a. User asks without naming or having discussed a procedure. ← 1. System asks which procedure or ministry the user needs. → 2. User names it, and the flow continues at step 3.

Use Case UC-11: RefreshOfficialContent	
Related Requirements	REQ-4, REQ-8, REQ-29, REQ-30
Initiating Actor	Scheduler (System Timer)
Actor's Goal	To keep the stored government information current, so that answers and last-verified dates reflect what ministries have most recently published.
Participating Actors	Official Sources (Ministry Websites, Facebook Pages, Telegram Channels)
Preconditions	Each covered ministry's official website, public Facebook page, and public Telegram channel is registered in SuperAsk.AI's source list with its link and platform.
Postconditions	New or changed content is stored with its source link and fetch date, the last-verified date of every source checked successfully is updated, and a refresh report records which sources were checked, changed, or failed.

Flow of Events for Main Success Scenario	→ 1. Scheduler starts the content refresh at its set time. ← 2. System loads the list of registered official sources. ← 3. System requests the latest page or posts from each Official Source: a ministry web page, a public Facebook page, or a public Telegram channel. → 4. Official Source returns its content. ← 5. System extracts the text, removing menus and other repeated page elements. ← 6. System compares the text with the stored copy. ← 7. System saves any new or changed text with its source link and fetch date, and updates the information used for answers. ← 8. System updates the last-verified date of each source it checked successfully. ← 9. System records a refresh report listing the sources checked, changed, and failed.
Flow of Events for Extensions (Alternate Scenarios)	2a. A new ministry's sources were added to the source list. ← 1. System includes them in this refresh, with no change to the answer-generation logic (REQ-30). 3a. The source's robots.txt rules do not allow automated access. ← 1. System skips the source and records it as blocked. 4a. The source does not respond or returns an error. ← 1. System keeps the stored copy and its previous last-verified date, and records the failure. 4b. The source returns a login or block page instead of real content. ← 1. System treats the response as a failure, not as changed content. 6a. The content has not changed. ← 1. System keeps the stored text and only updates the last-verified date.

d. **System Sequence Diagrams**

The system sequence diagrams below show the messages exchanged between the actors and SuperAsk.AI, which is treated as a single black box, for the five use cases elaborated in Section 3c-iv. Solid arrows are requests, dashed arrows are responses, and the orange frames mark alternative (alt), optional (opt), and repeated (loop) parts of the flow.

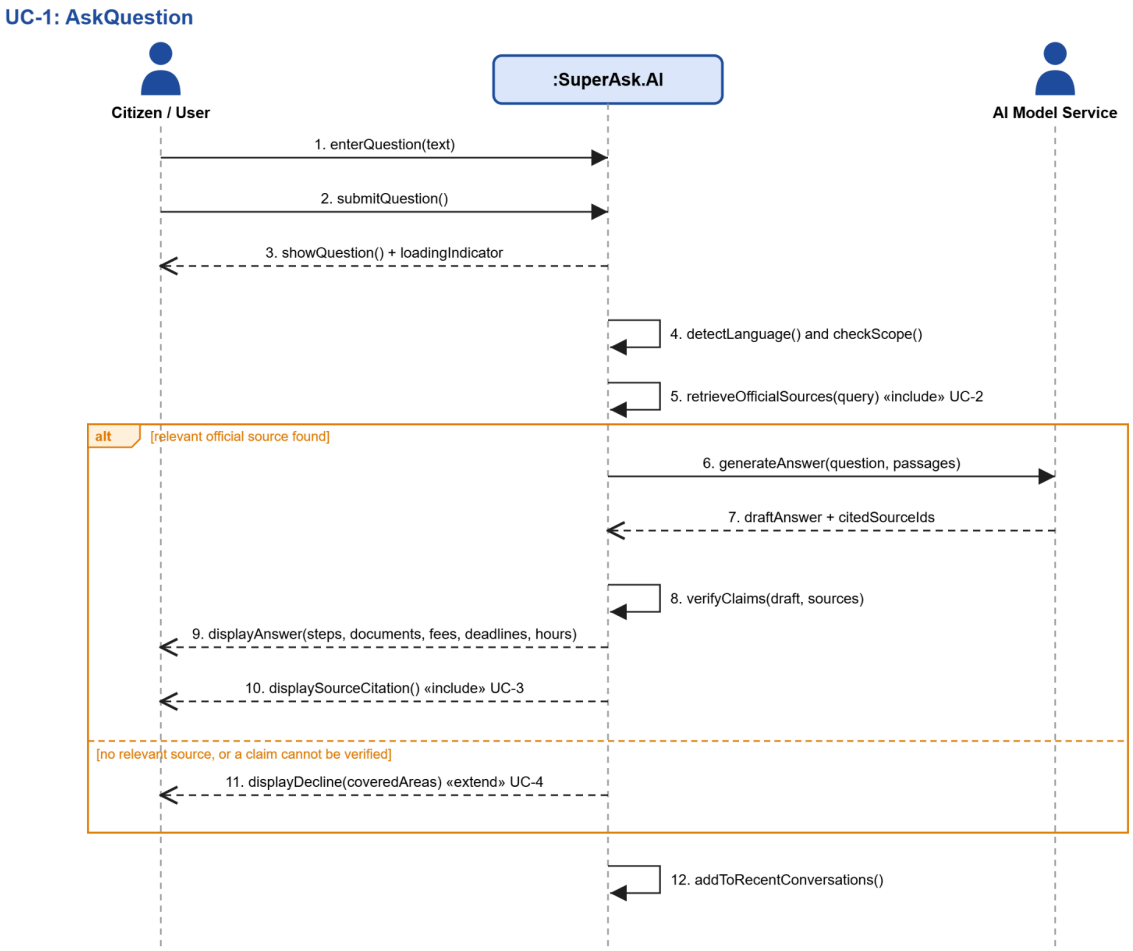


Figure 3: System sequence diagram for UC-1: AskQuestion

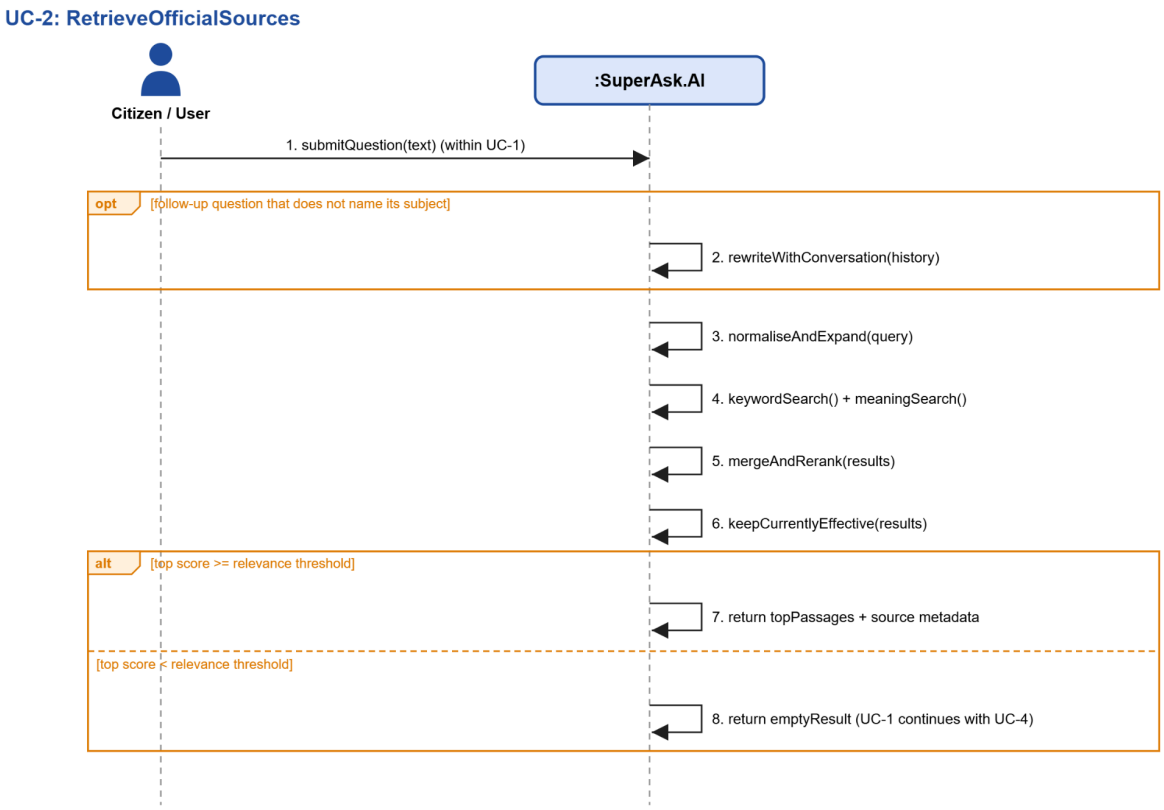


Figure 4: System sequence diagram for UC-2: RetrieveOfficialSources



Figure 5: System sequence diagram for UC-3: ViewSourceCitation

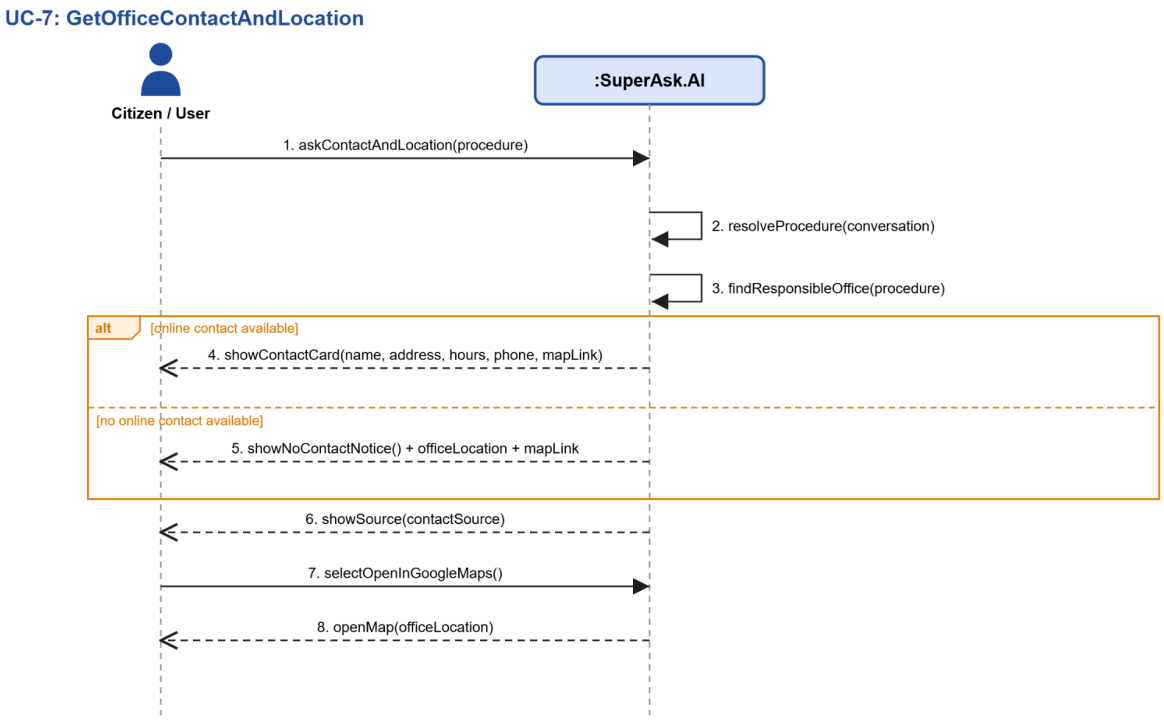


Figure 6: System sequence diagram for UC-7: GetOfficeContactAndLocation

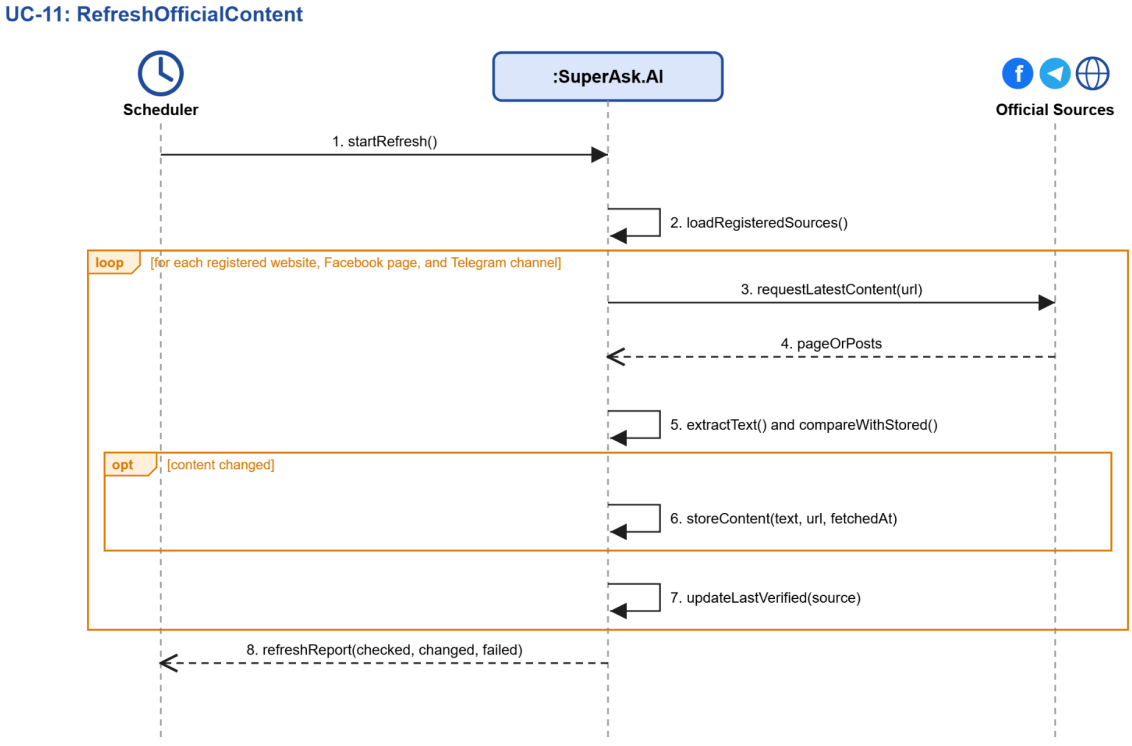


Figure 7: System sequence diagram for UC-11: RefreshOfficialContent

4. User Interface Specification

a. Preliminary Design

The SuperAsk.AI user interface is designed to provide citizens with a simple way to access information about Cambodian government services and procedures. The main interaction uses a chat-style interface so that users can ask questions in natural language rather than searching through multiple government websites or documents.

The interface design follows the requirements established in Report One. Users can enter and submit questions, view generated answers and verified official sources, select commonly requested procedures, return to recent questions, and use the interface in Khmer or English. The following mock-ups show the main navigation path, the information entered by the user, the controls used to complete each interaction, and the information displayed by SuperAsk.AI.

1. Starting a Government-Service Question

The user can access the chat interface from the SuperAsk.AI Home Page by selecting Ask on the web or Ask now. The Home Page also provides access to language selection and supporting information about the service.

Navigation Path:
Home Page → Ask on the web / Ask now → Chat Interface → Enter Question or Select Procedure → Send / Select → Response

User Interaction:

1. The user opens the SuperAsk.AI Home Page.
2. If necessary, the user selects Khmer or English from the language control.
3. The user selects Ask on the web or Ask now to enter the chat interface.

Primary controls: Ask on the web, Ask now, language selection
Result: The user enters the SuperAsk.AI chat interface.

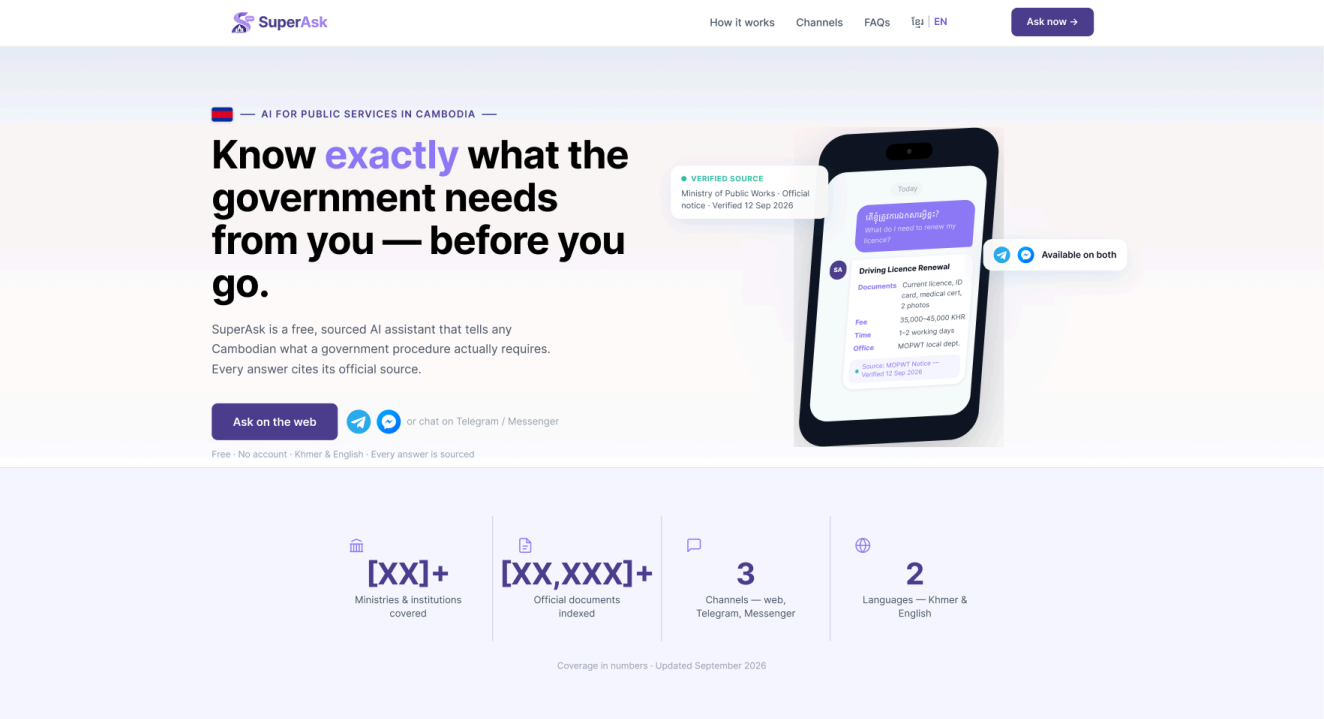


Figure 8. SuperAsk.AI Home Page

The initial chat screen provides three ways to begin an interaction. The user can enter a new question in the input field, select a previous question from Recent, or choose one of the Popular Procedures displayed in the interface.

4. The user may select a question from Recent, select a procedure from Popular Procedures, or use the main question input field.
5. To ask a new question, the user clicks the input field.
6. The user types a government-service question in Khmer or English.
7. The user presses the Send button.
8. SuperAsk.AI displays the corresponding response in the conversation area.
9. The user can continue asking follow-up questions through the same input field.

Input field: Government-service question

Primary controls: Send button, Recent questions, Popular Procedures, New chat, language control

Displayed result: The submitted question and corresponding response

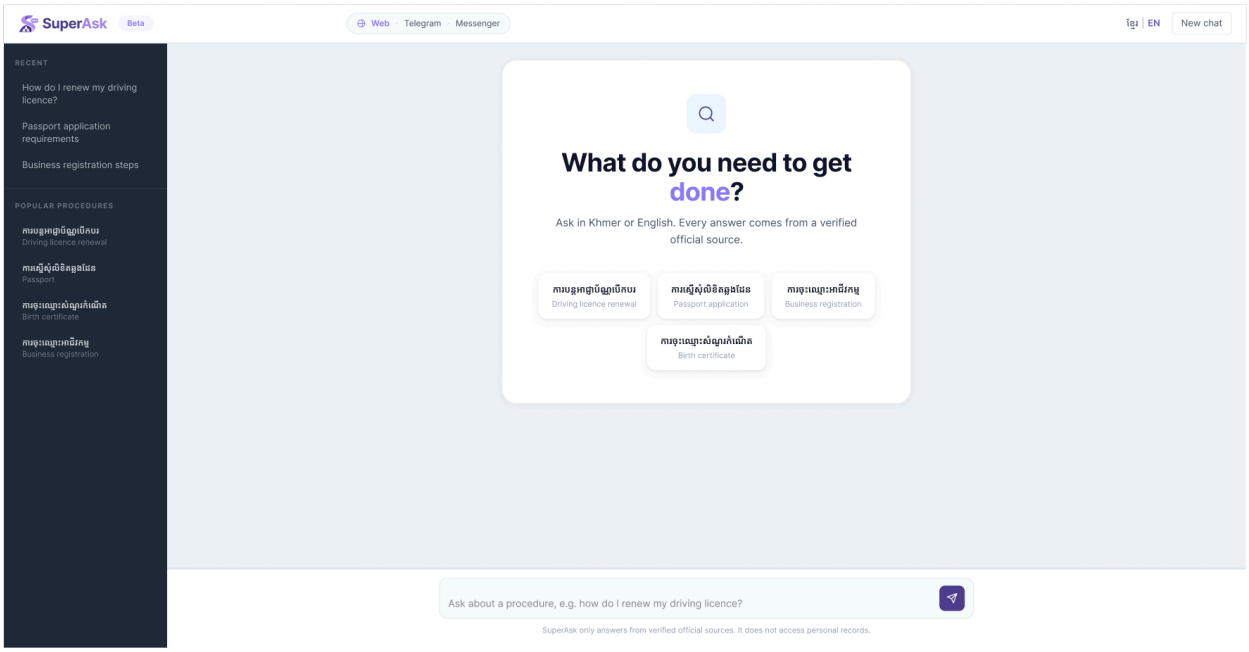


Figure 9. SuperAsk.AI initial chat interface

2. Viewing a Government-Procedure Answer

When SuperAsk.AI finds relevant information from a verified government source, the answer is displayed as a structured response card. The current prototype uses a driving licence renewal question to demonstrate this interaction.

The user's submitted question is displayed in a separate message bubble. The answer card then identifies the responsible government organization and presents the important procedure information in separate sections.

For the procedure shown in the mock-up, the response layout contains fields for Documents, Fee, Time, and Office, followed by the required procedure Steps. The verified source is displayed in a separate source area beneath the answer.

Navigation Path:
Chat Interface → Enter Question → Send → Procedure Answer → View Verified Source / Office Information

User Interaction:

1. The user enters a government-procedure question in the input field.
2. The user presses the Send button.
3. The submitted question appears in the conversation area.
4. SuperAsk.AI displays the relevant procedure answer.

5. The user reviews the required documents, fee, expected processing time, responsible office, and procedure steps when that information is available.
6. The user reviews the Verified Source displayed beneath the procedure information.
7. The user may select the source-link control to access the referenced source.
8. If location information is available, the user may select Show office on map.
9. The user may select Yes or No under Was this helpful? to provide feedback.
10. The user can enter a follow-up question through the input field or select New chat to begin another conversation.

Input: Government-procedure question
Primary controls: Send, verified-source link, Show office on map, Yes/No feedback, New chat
Displayed result: Procedure information, required documents, fees, processing time, responsible office, ordered steps, and verified source

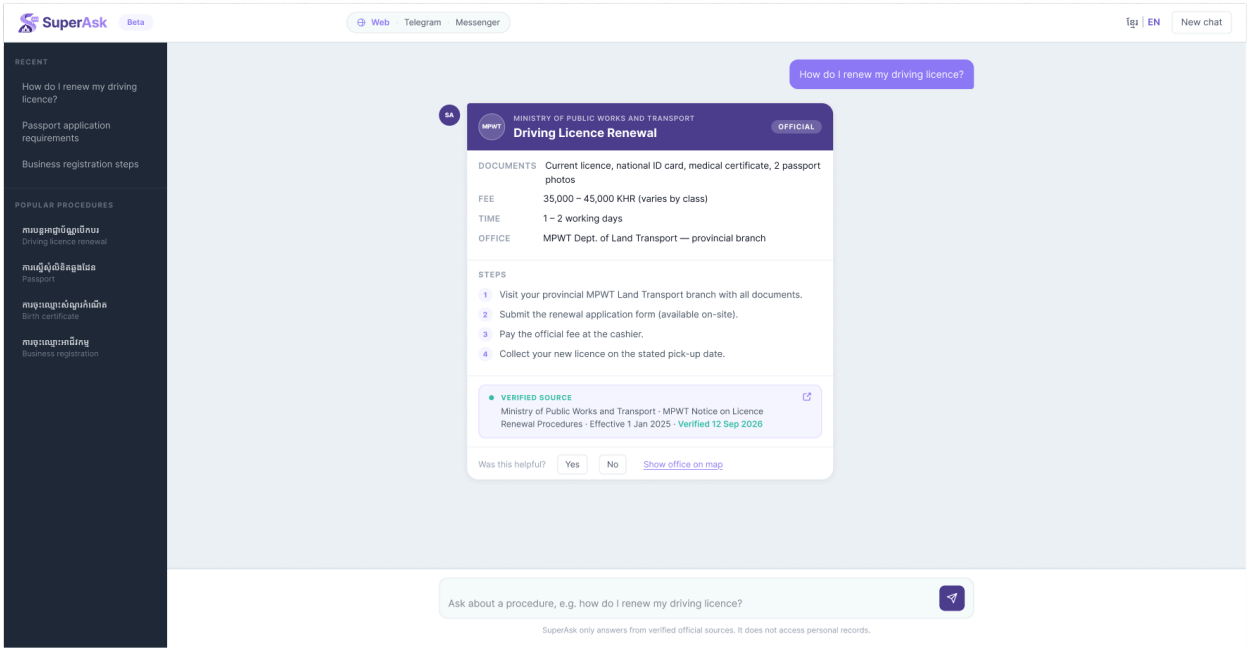


Figure 10. SuperAsk.AI government-procedure answer with verified source.

3. Handling an Unsupported or Out-of-Scope Question

SuperAsk.AI is specifically designed for questions related to Cambodian government procedures. Therefore, the interface also needs to clearly handle questions that fall outside the supported scope.

The out-of-scope mock-up demonstrates a user submitting the message *"hello."* Instead of generating an unrelated response, SuperAsk.AI displays a message explaining that the

service only answers questions about Cambodian government procedures and suggests relevant types of questions.

Navigation Path:
Chat Interface → Enter Unsupported Question → Send → Out-of-Scope Message → Enter Another Question

User Interaction:

1. The user types a question or message into the chat input field.
2. The user presses the Send button.
3. The submitted message appears in the conversation area.
4. If the request is outside the supported scope, SuperAsk.AI displays an out-of-scope response instead of presenting an unsupported answer.
5. The response explains that SuperAsk.AI is intended for Cambodian government procedures.
6. The interface suggests examples of supported topics, such as licences, certificates, registrations, or permits.
7. The user may return to the input field and enter another question.
8. Alternatively, the user may select New chat to begin a new conversation.

Input: Unsupported or unrelated question
Primary controls: Send button, question input, New chat
Displayed result: Out-of-scope guidance message

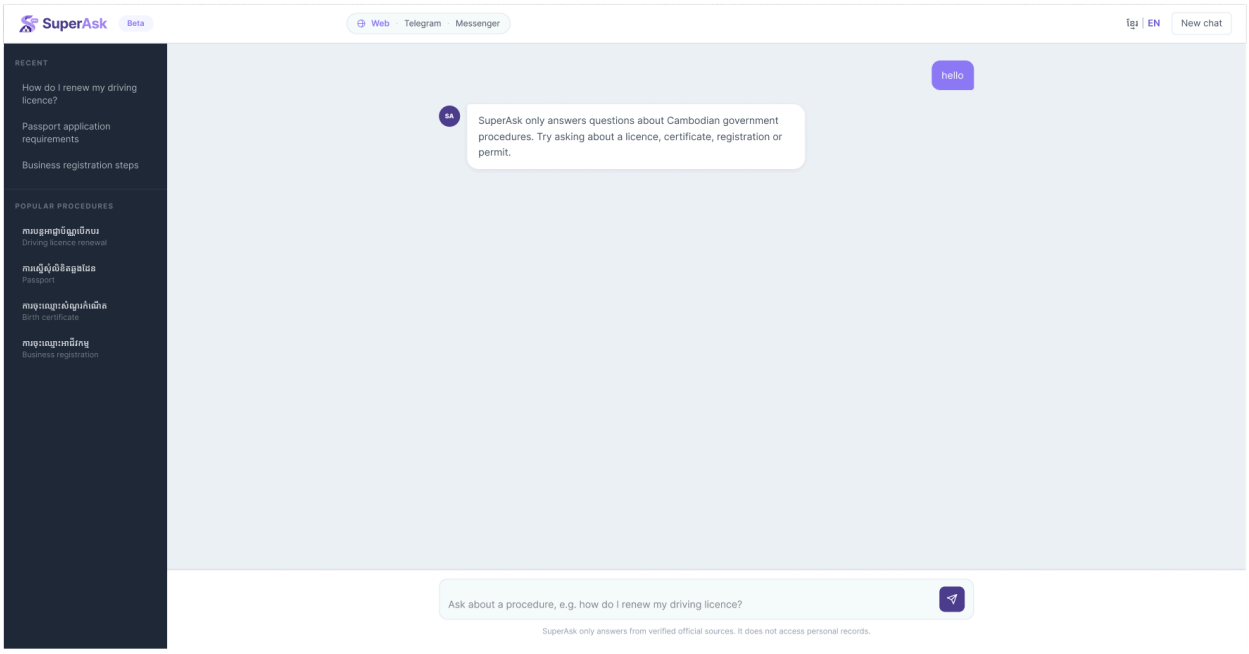


Figure 11. SuperAsk.AI out-of-scope response.

The interface was designed to minimize unnecessary navigation. Users can ask a question directly instead of first selecting a ministry or category, while popular procedures remain available as optional shortcuts. Procedure answers use a structured card rather than a long paragraph so that documents, fees, processing time, office information, steps, and verified sources are easier to identify.

b. User Effort Estimation

1. Ask a Government Service Question

The following sequence of actions allows a user to enter and submit the government-service question “How do I renew my driving licence?” through the Superask.AI chat interface.

Navigation: Total of 1 click, as follows:

1.1. Click the chat input box.

Data Entry: Total of 1 click and 34 key presses, as follows:

1.2. Press 34 keys to enter “How do I renew my driving license?”
1.3. Click the Send button to submit the question.

2. Select a Suggested Procedure

The following sequence of actions allows a user to select one of the suggested government-service procedures displayed by SuperAsk.AI instead of manually entering a question.

Navigation: Total of 1 click, as follows:

2.1. Click one of the suggested procedure options.

Data Entry: Total of 0 key presses.

The system displays the corresponding procedure information without requiring the user to manually type a question.

3. View a Verified Source

The following sequence of actions allows a user to access the official source associated with a response.

Navigation: Total of 1 click, as follows:

3.1. Click the Source link displayed below the answer.

Data Entry: Total of 0 key presses.

The system opens the verified source associated with the government-service information.

4. **View an Office Location**

The following sequence of actions allows a user to view the location of the government office associated with a procedure.

Navigation: Total of 1 click, as follows:

4.1. Click the Show on Map button.

Data Entry: Total 0 key presses.

The system displays the relevant office location on the map.

5. **Ask a Follow-Up Question**

The following sequence of actions allows a user to ask a follow-up question after receiving a government-procedure response. For this example, the user asks “How long does it take?”

Navigation: Total of 1 click, as follows:

5.1. Click the chat input box.

Data Entry: Total of 1 click and 22 key presses, as follows:

5.2. Press 22 keys to enter “How long does it take?”
5.3. Click the Send button to submit the follow-up question.

6. **Start a New Chat**

The following sequence of actions allows the user to clear the current conversation and begin another request.

Navigation: Total of 1 click, as follows:

6.1. Click the New Chat button.

Data Entry: Total of 0 key presses.

The system opens a new chat session where the user can enter another government-service question.

5. Project Management

Plan of Work

Following the submission of Report One, our team will continue developing the system through two main iterations. The first iteration will focus on developing and integrating the core functions needed for the first demonstration, including the web interface, bilingual question handling, retrieval from verified government sources, citation display, and guardrails for unsupported questions. The second iteration will focus on refinement, additional ministry coverage, testing, and improvements based on feedback from the first demonstration. The short-term activities are described in greater detail because they are the immediate priorities of the team. Later milestones may be adjusted based on technical progress, testing results, and instructor feedback.

Period	Planned Work	Milestone
Sep 21 - 25, 2026	Complete customer requirements, business goals, functional/nonfunctional requirements, UI requirements, project management, and preliminary interface designs.	Report #1 Parts 1 & 2
Sep 26 - 30, 2026	Review and revise requirements; ensure consistency between the problem statement, requirements, prototype, and team responsibilities.	Full Report #1 – Sep 30
Oct 1 - 6, 2026	Continue collecting and verifying official government information. Develop interaction diagrams and begin implementing the retrieval and answer-generation workflow.	Report #2 Part 1 – Oct 6
Oct 7 - 12, 2026	Develop class/system architecture diagrams. Continue development of the web interface, database, retrieval system, and bilingual processing.	Report #2 Part 2 – Oct 12
Oct 13 - 21, 2026	Integrate the interface, retrieval pipeline, citation system, guardrails, and verified-source database. Conduct initial integration testing.	Full Report #2 – Oct 21
Oct 22 - Nov 1, 2026	Implement the core use cases selected for the first demonstration. Test Khmer and English queries and verify responses against official sources.	First-demo implementation

Nov 2–8, 2026	Debug the integrated system, test unsupported/out-of-scope questions, improve response quality, and prepare the demonstration.	First-demo preparation
Nov 9, 2026	Demonstrate the first working iteration and collect feedback.	Demo #1
Nov 10–16, 2026	Apply feedback from Demo #1, expand verified government information, improve interface behavior, and conduct additional system testing.	Report #3 Part 1 – Nov 16
Nov 17–23, 2026	Complete system testing and final documentation. Verify consistency between implementation, requirements, and reports.	Full Report #3 – Nov 23
Nov 24–Dec 3, 2026	Refine the system, resolve remaining defects, improve ministry coverage, and prepare the final demonstration.	Final refinement
Dec 4, 2026	Present the completed SuperAsk.AI system.	Demo #2

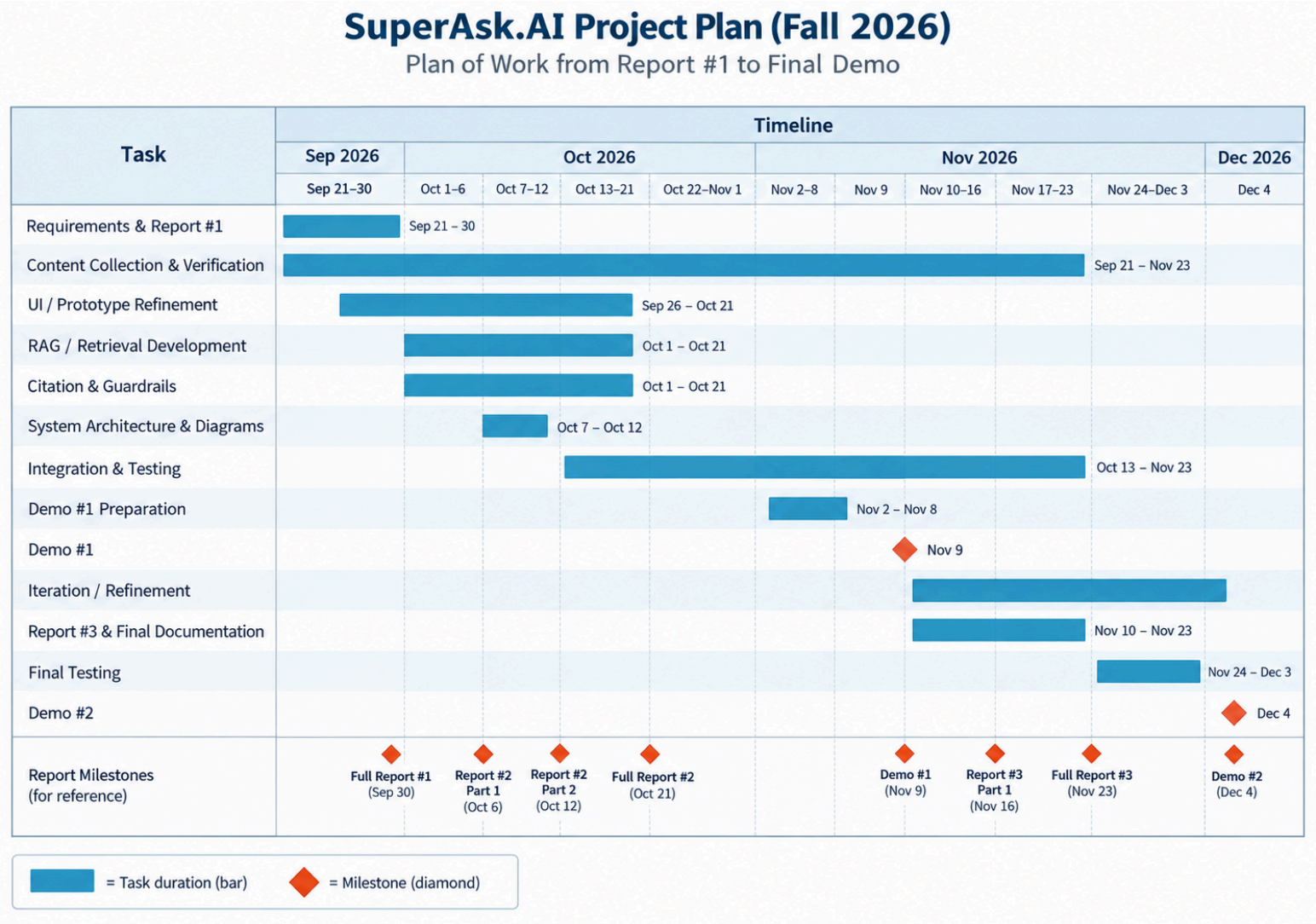


Figure 12: Project Plan

The Gantt chart summarizes the planned development activities and major project milestones from Report #1 through the final demonstration.

Product Ownership and Team Responsibilities

SuperAsk.AI is being developed collaboratively by six team members. Each member has a primary project role based on their technical and organizational strengths, while major design, testing, and implementation decisions are discussed collectively. The Project Lead coordinates work between the different roles and monitors progress against the semester schedule.

Team Member	Role	Work Completed / So Far	Current Responsibilities	Future Responsibilities
Yean Fong	**Project Lead**	Coordinated the project proposal, project scope, team organization, scheduling, and requirements preparation. Contributed to design and implementation documentation.	Coordinate Report #1, requirements, deadlines, communication between team members, and overall project consistency.	Monitor milestones, coordinate system integration and demos, assist testing/documentation, and manage final project delivery.
Phallamonunhh Chun	**AI/ML Engineer**	Contributed to business analysis, AI planning, cybersecurity considerations, and the SuperAsk.AI concept.	Develop and evaluate the RAG pipeline, model integration, retrieval behavior, and AI guardrails.	Improve answer accuracy, retrieval performance, refusal behavior, and bilingual AI handling.
Bunkheang Heng	**Web/API Developer**	Contributed technical development, system architecture planning, full-stack concepts, and database planning.	Develop the web application, APIs, database connections, and integration with the AI system.	Complete integration, optimize backend functions, fix technical defects, and support deployment.
Doungkhe Neang	**Data/Content Curator**	Contributed data organization, content planning, documentation, and source research.	Collect, organize, verify, and categorize official government content used by the system.	Expand ministry coverage, maintain source metadata, and help ensure information remains current and properly cited.

Sereyprophea Sarin	**QA/Testing Lead**	Contributed documentation, communication, interface discussion, and initial testing planning.	Develop test questions and check accuracy, bilingual responses, source citation, and refusal behavior.	Conduct integration and user testing, record defects, collect user feedback, and verify corrections before demonstrations.
Heejae Choi	**UX/UI Designer**	Contributed interface planning, UX/UI concepts, page layout, and prototype development.	Refine user flows, chat interface, responsive layouts, and presentation of answers and citations.	Improve the interface based on user testing and ensure consistency across desktop/mobile views and supported channels.

Management and Coordination

The Project Lead will maintain the overall project schedule and coordinate dependencies between the sub-teams. Development tasks will be reviewed regularly to identify delays or technical problems early. Team members remain primarily responsible for their assigned areas, but implementation and testing will overlap when integration requires cooperation between AI, content, interface, backend, and QA work.

Team Member	Competencies	Assigned Area
Yean Fong	Project management, organizational skills, assertive communication, Python, C++, JavaScript, HTML, CSS, Design	**Management / Integration**
Phallamonunhh Chun	UX/UI, Business Analysis, Cybersecurity, CSS, C++, AI	**AI / Data**
Bunkheang Heng	Technical development, System architecture, Full Stack Development, Database Design, Machine Learning	**Development**

Doungkhe Neang	UX/UI, data curation and management, technical documentation, information organization	**AI / Data**
Heejae Choi	UX/UI, interface planning, C++, JavaScript, HTML, CSS, documentation	**UX / UI**
SereyProphea Sarin	Documentation, communication, UX/UI, Java, JavaScript, HTML, CSS	**Testing / QA**

References

SuperAsk.AI Team AO3. (2026). ***SuperAsk.AI GitHub repository*****.**
https://github.com/Monunhh/Super-ask.git
Used for project source code, version control, and team development.

Open Government Products. (2026). ***AskGov Singapore*****. Singapore Government.**
https://ask.gov.sg/
Used as a reference for how government information can be organized by agency and presented through a searchable question-and-answer interface. AskGov lists government organizations and public-service questions in one place.